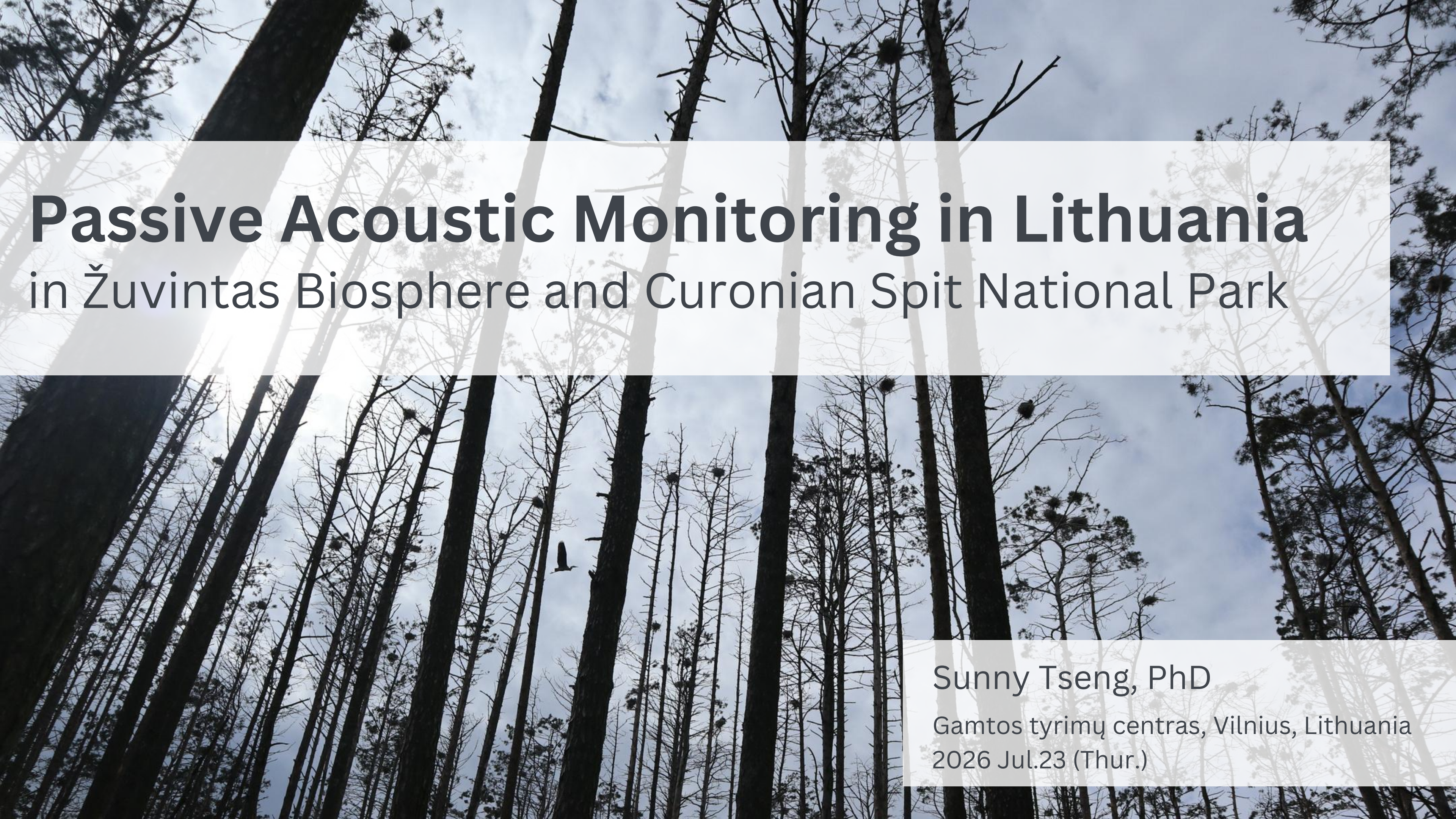




Passive Acoustic Monitoring in Lithuania

in Žuvintas Biosphere and Curonian Spit National Park

Sunny Tseng, PhD

Gamtos tyrimų centras, Vilnius, Lithuania
2026 Jul.23 (Thur.)

Sveiki!



2017 - present

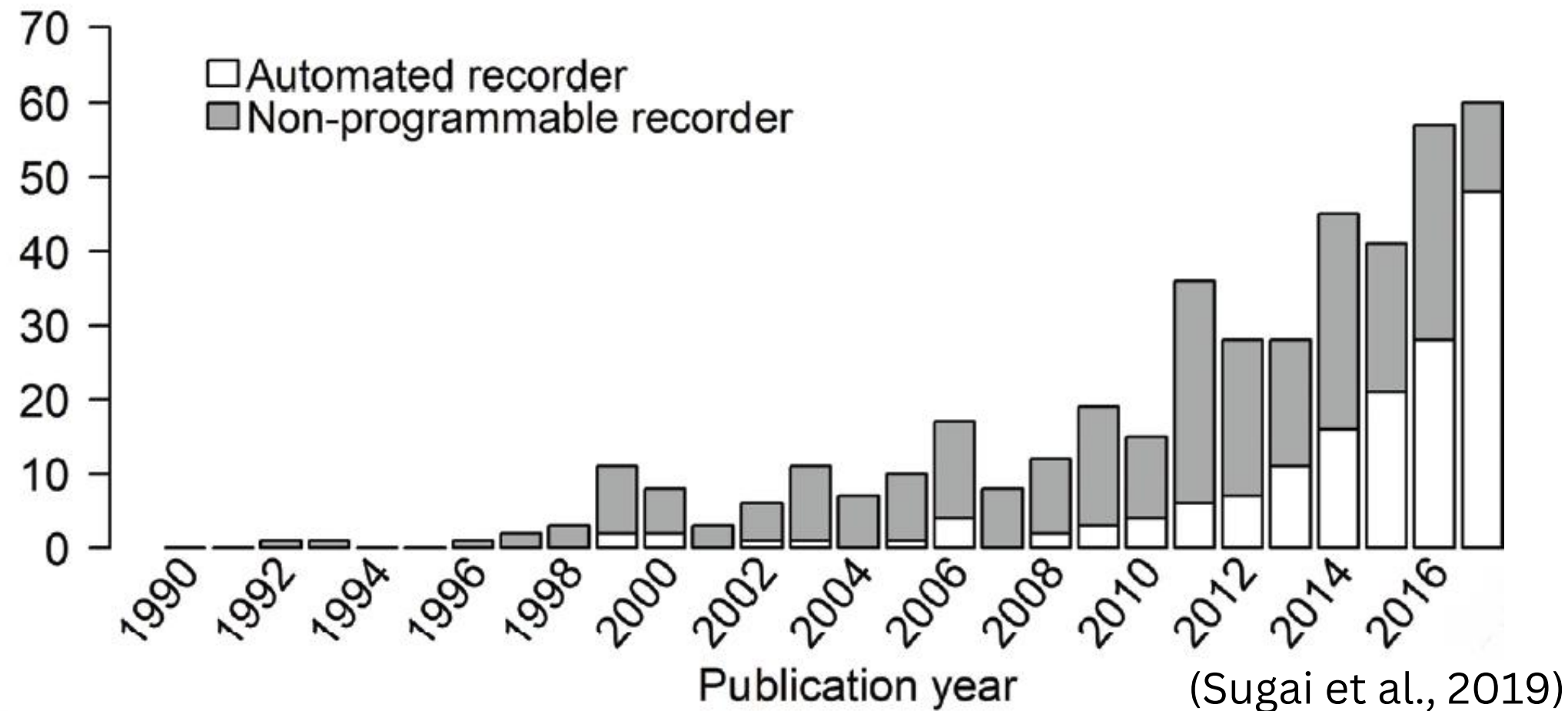
here!



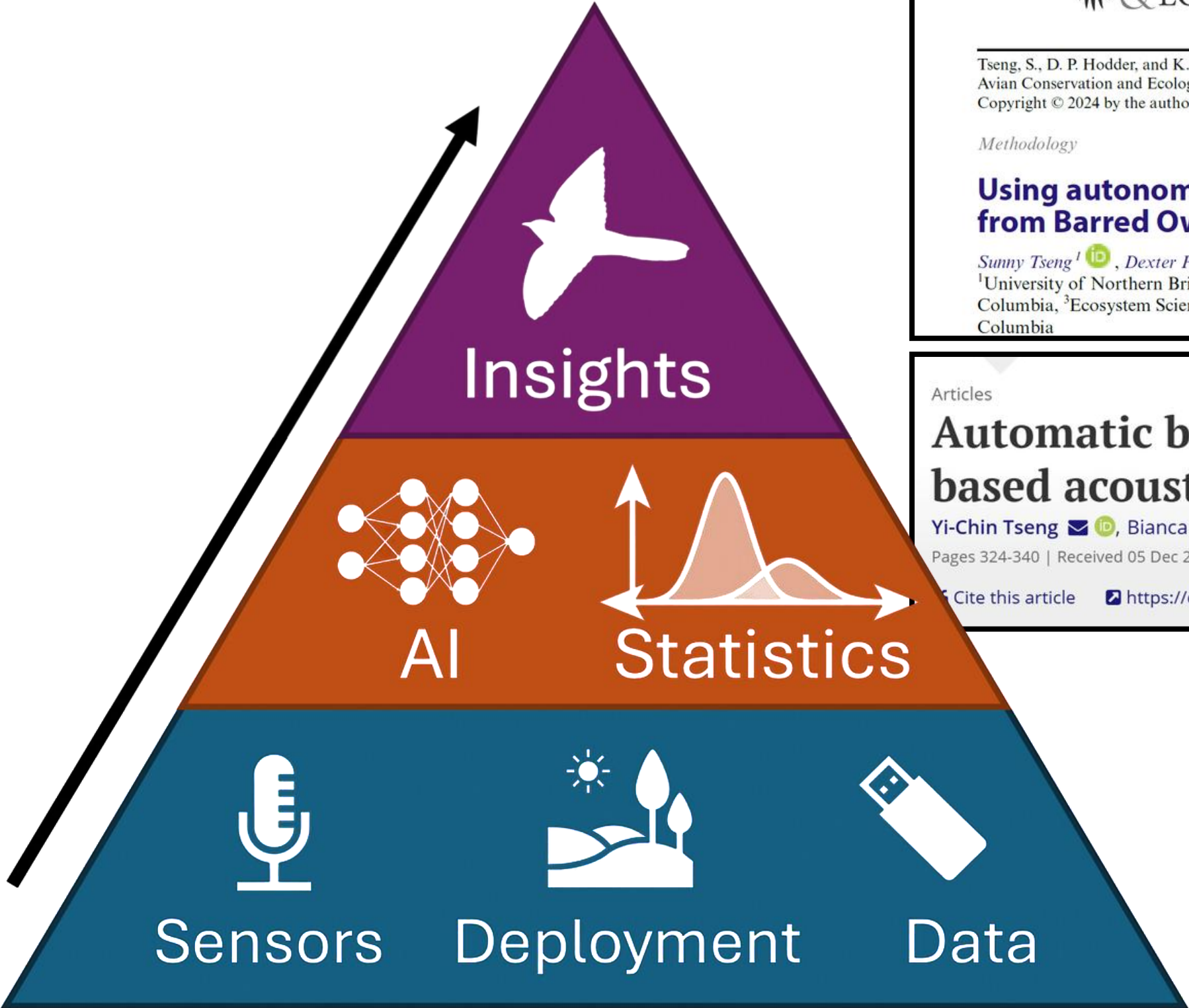
Sveiki!

- BSc. Physics & Forestry conservation, Taiwan
- PhD., MSc in British Columbia, Canada
- Passive acoustic monitoring
- Machine learning

“passive acoustic monitoring” published articles in past decade



Passive acoustic monitoring in avian ecology



 AVIAN CONSERVATION & ECOLOGY

VOLUME 19, ISSUE 1, ARTICLE 23

Tseng, S., D. P. Hodder, and K. A. Otter. 2024. Using autonomous recording units for vocal individuality: insights from Barred Owl identification. Avian Conservation and Ecology 19(1):23. <https://doi.org/10.5751/ACE-02680-190123>
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Methodology

Using autonomous recording units for vocal individuality: insights from Barred Owl identification

Sunny Tseng¹ , Dexter P. Hodder²  and Ken A. Otter³

¹University of Northern British Columbia, Prince George, British Columbia, ²John Prince Research Forest, Fort. St. James, British Columbia, ³Ecosystem Science & Management Program, University of Northern British Columbia, Prince George, British Columbia

Articles

Automatic bird sound detection: logistic regression based acoustic occupancy model

Yi-Chin Tseng  , Bianca N. I. Eskelson, Kathy Martin & Valerie LeMay

Pages 324-340 | Received 05 Dec 2019, Accepted 11 Feb 2020, Published online: 24 Feb 2020

Cite this article  <https://doi.org/10.1007/s10336-025-02260-w>

Journal of Ornithology
<https://doi.org/10.1007/s10336-025-02260-w>

METHODS PAPER

Setting BirdNET confidence thresholds: species-specific vs. universal approaches

Sunny Tseng¹  · Dexter P. Hodder² · Ken A. Otter¹

Received: 7 August 2024 / Revised: 12 December 2024 / Accepted: 4 February 2025
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(Tseng & Kahl, 2026, under review)

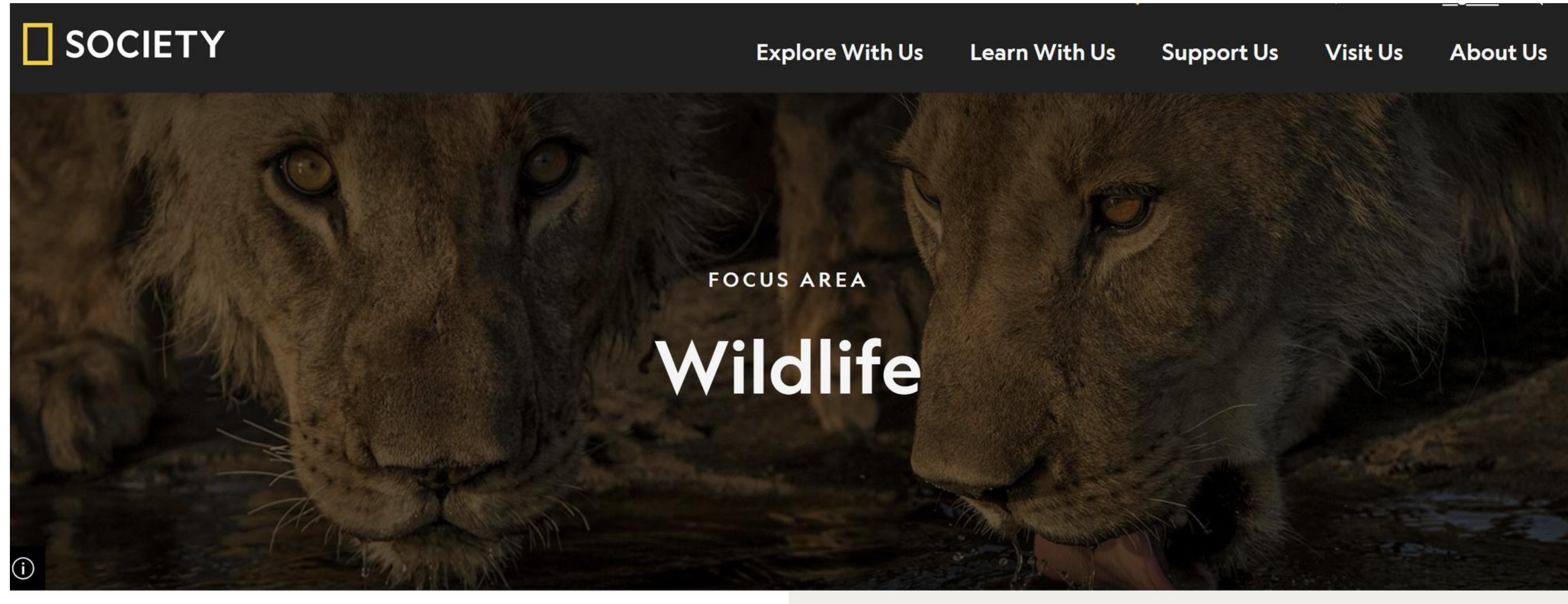
(Tseng et al., 2024)

(Tseng et al., 2020)

(Tseng et al., 2025)

(Kitzes et al., 2026)

Baseline audio dataset - Early Career Grant



- Association of species composition with habitat
- Species vocalization behaviour across seasons
- Between species interaction
- ...etc.

A global PAM (meta-)database

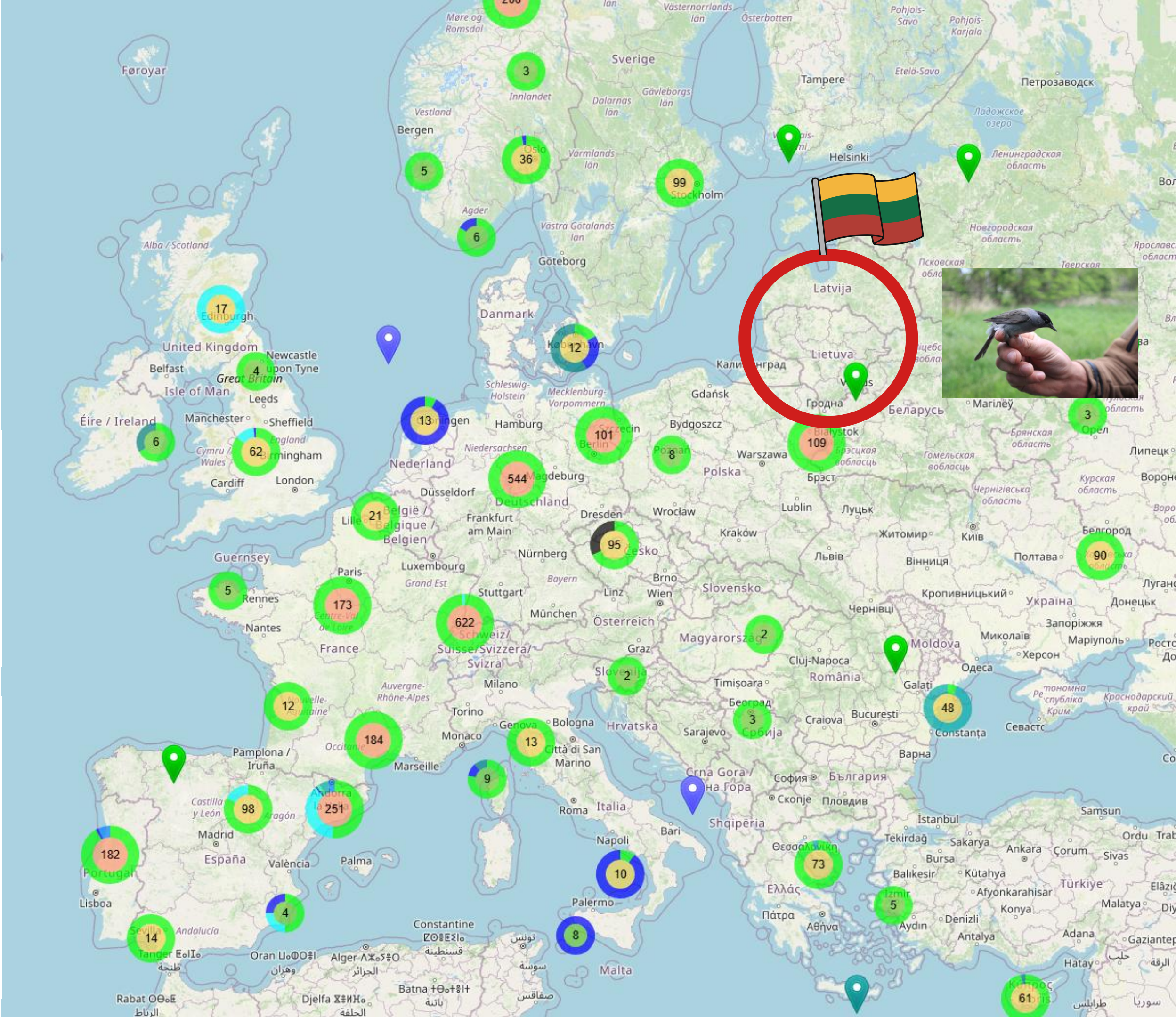
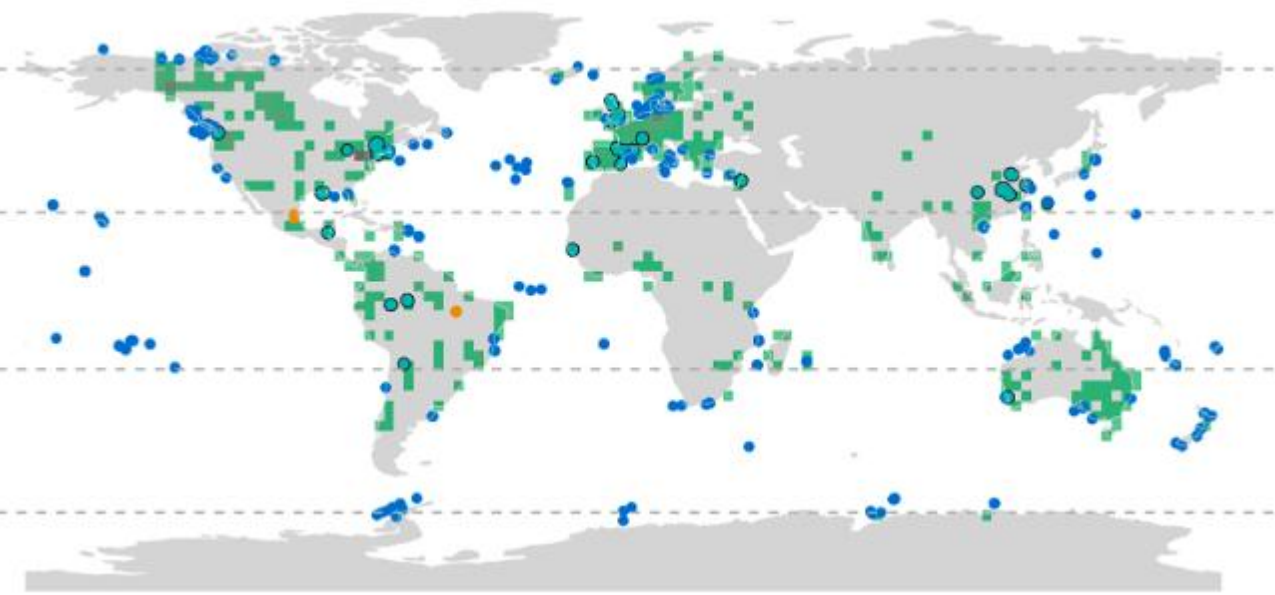
Global Ecology and Biogeography

A Journal of Macroecology

RESEARCH ARTICLE | Open Access | CC BY

Worldwide Soundscapes: A Synthesis of Passive Acoustic Monitoring Across Realms

Kevin F. A. Darras , Rodney A. Rountree, Steven L. Van Wilgenburg, Anna F. Cord, Frederik Pitz, Youfang Chen, Lijun Dong, Agnès Rocquencourt, Camille Desjonquères ... See all authors



Xeno-Canto - world bird sound library

xeno-canto

Sharing wildlife sounds from around the world

Search recordings...

Search

Advanced Search Tips

About

Explore

Upload

Forum

Mysteries

Articles

Logged in as Sunny Tseng

"Lost birds" on XC

Upload Sounds

Upload Annotations



New Ireland Friarbird, New Ireland. © Danny Bregman

XC1148089

0:00

1:33

New Ireland Friarbird *Philemon eichhorni*

Cornel Schepers

In a new article

John Mittermeijer explains some on the so-called "lost" birds project of the American Bird Conservancy, and of the role that contributions to Xeno-canto can play in it. Check it out. This spotlight is illustrated with an image and recording of New Ireland Friarbird, one of the "lost" birds found recently, and one that had never been sound recorded before.

All groups

Collection Statistics

- 1074619 Recordings
- 1045091 IDs
- 922060 Background IDs
- 13351 Species
- 13066 Subspecies
- 12736 Recordists
- 19585:53:50 Recording Time
- 962135

More...

Latest News

Sri Lankan Chirixalus

Recordings

488 results from 97 species for query 'cnt:lithuania' (foreground species only)

Export all results as a CSV (spreadsheet) file (maximum 1000 results)

Make a map of all locations of the current selection

Add all results to a recording set

Results format: detailed | concise | sonograms

1

2

3

4

5

6

7

8

9

...

17

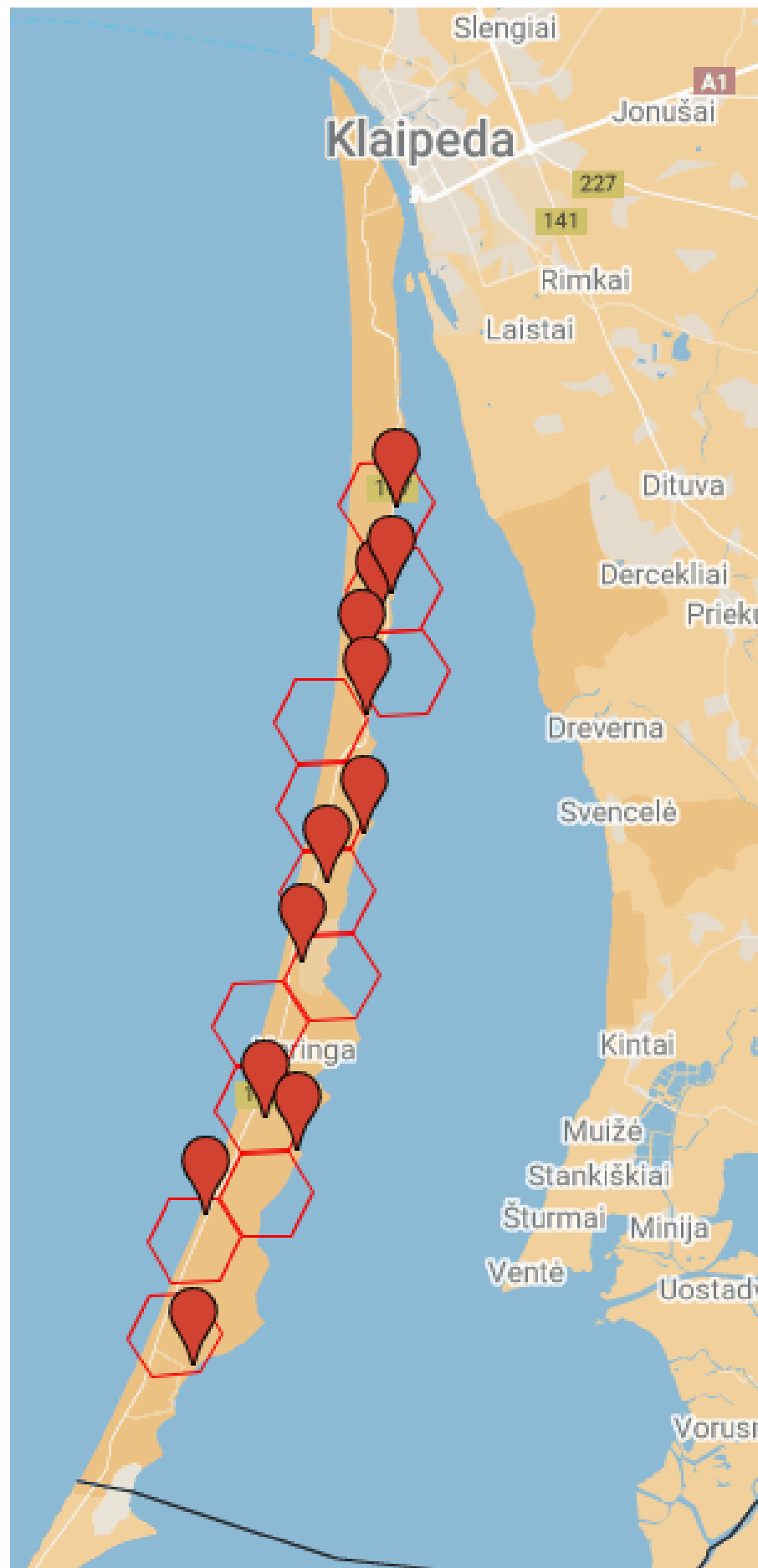
Next >

Common name / Scientific	Length	Recordist	Date	Time	Country	Location	Elev. (m)	Type (predef. / other)	Remarks	Actions / Quality	Cat.nr.
(?) Identity unknown (bird)	0:23	darelas	2016-04-02	14:30	Lithuania	Verkių seniūnija (near Vilnius), Vilniaus miesto savivaldybė, Vilniaus apskritis	130	song	very annoying theme tune. A meadow bird... more » Mystery recording. See the forum. [sono]		XC390693
Great Spotted Woodpecker <i>Dendrocopos major</i>	0:30	Sunny Tseng	2025-05-11	12:00	Lithuania	Juodkrantė (sound catcher)	50	drumming	Woodpecker sp. Same individual as XC1152... more » Originally a mystery recording. See the forum. [also] [sono]		XC1152683
(?) Identity unknown (bird)	0:07	Sunny Tseng	2025-05-11	07:30	Lithuania	Nida (reed area)	0	call	Audio processed by normalization (-3dB)... more » Mystery recording. See the forum. [sono]		XC1152619
European Robin <i>Erithacus rubecula</i>	0:28	Sunny Tseng	2025-05-08	08:30	Lithuania	Zuvintas Biosphere (nature trail)	110	call	Audio processed by high-pass filter (300... more » Originally a mystery recording. See the forum. [sono]		XC1152299
Mute Swan <i>Cygnus olor</i>	0:31	Sunny Tseng	2025-05-09	07:00	Lithuania	Zuvintas Biosphere	80	flight call	Saw the birds flying by, with swan... more » Originally a mystery recording. See the forum. [sono]		XC1152283

Latest News

Random Recording

81 species, 284 recordings



Curonian Spit National Park



sand dune



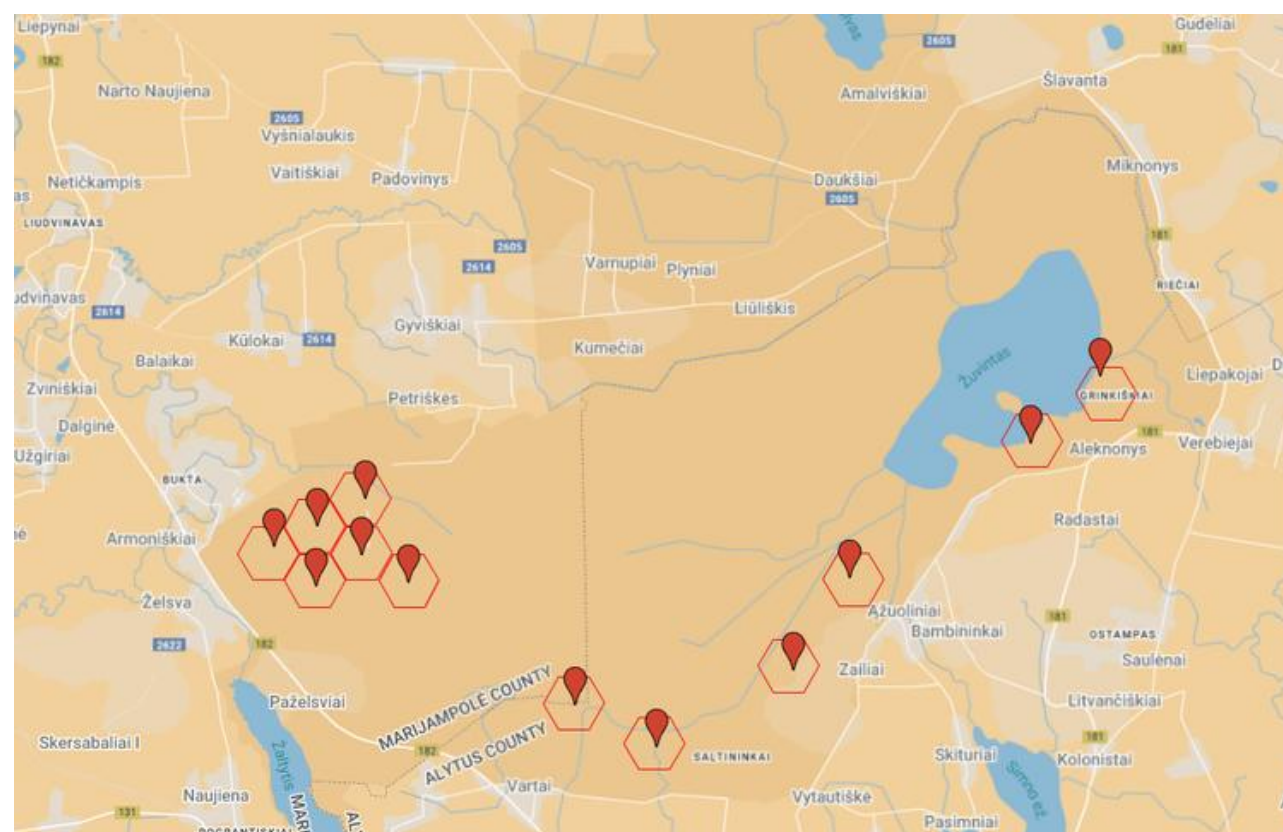
planted pine forest



reed



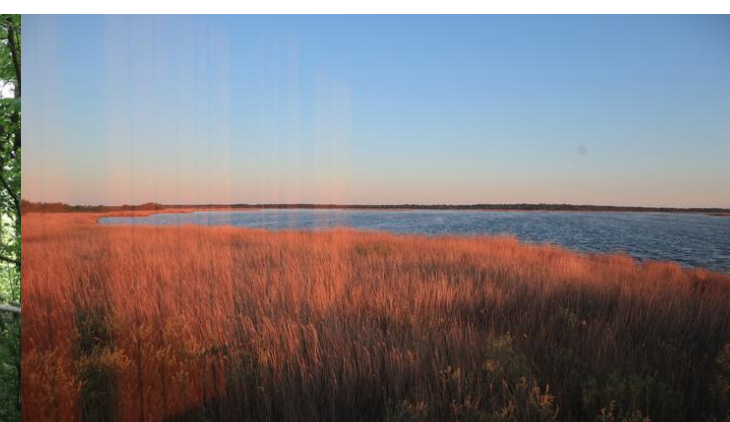
old-growth forest



Žuvintas Biosphere Reserve



forest



lakeside



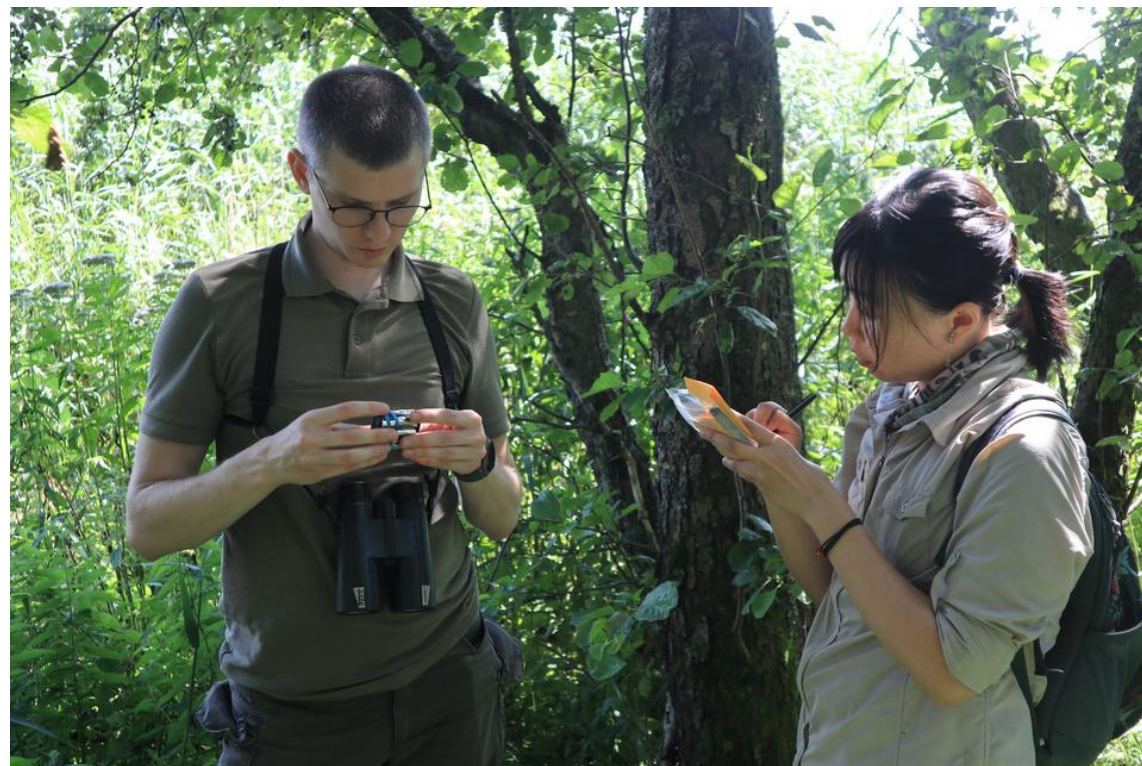
marsh



Fieldwork

2025 May - July

- 24 audio recording stations (24 hours)
- Parabolic mic for species-specific vocalization



Curonian Spit National Park
Modestas Bružas



Žuvintas Biosphere Reserve
Arūnas Pranaitis



Field equipment





Single species recording collection (Xeno-Canto)



Lithuania: Vilnius (150 m)



Taiwan: Zhongzhiguan (2,300 m)



Taiwan: Lalashan (1,600m)



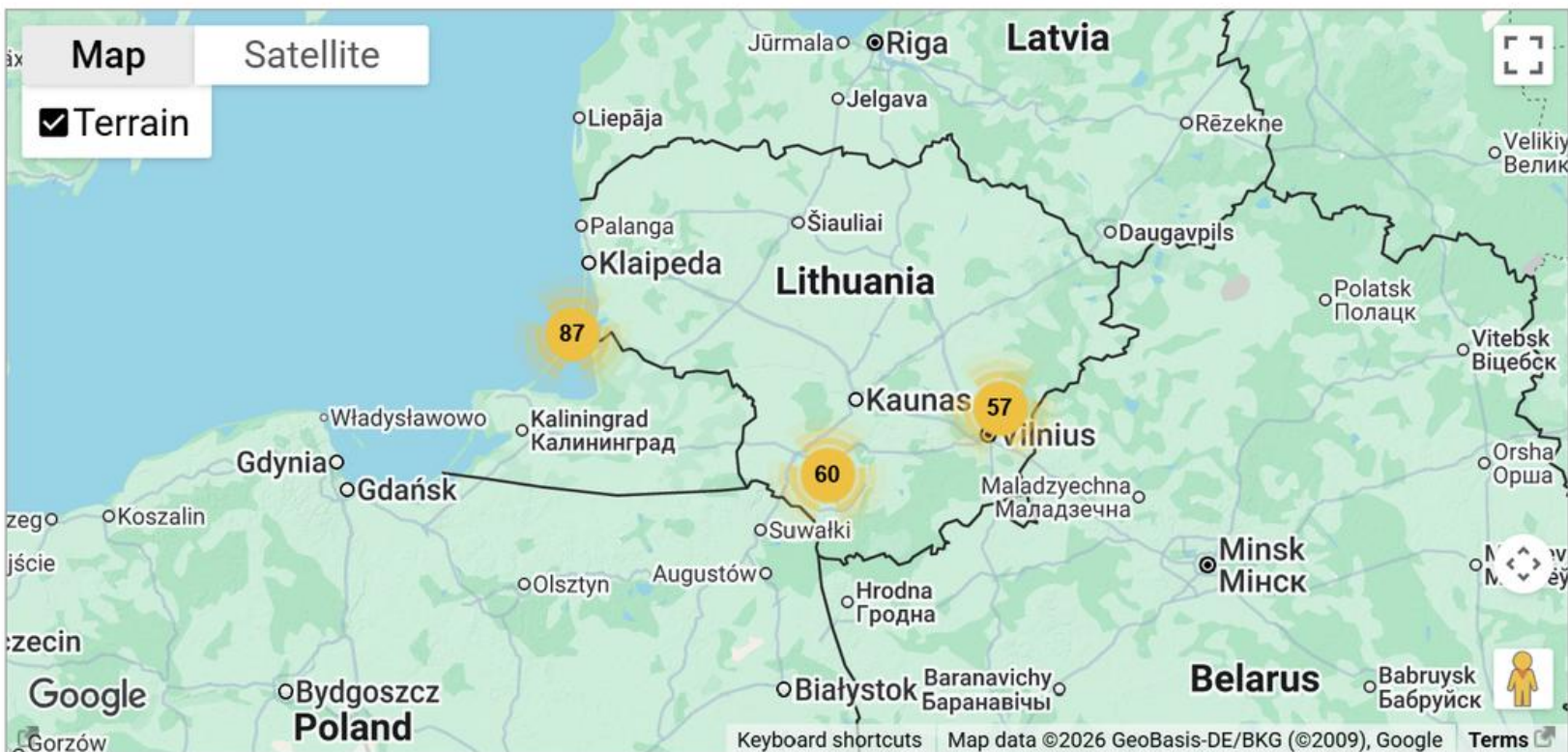
Lithuania: Juodkrantė



Taiwan: Mt. Sylvia



Lithuania: Zuvintas Biosphere



cnt: lithuania [Filter results](#)

Results format: **detailed** | concise | **sonograms**

1 2 3 4 5 6 7 Next >

	Common name / Scientific	Length	Recordist	Country	Location	Elev. (m)	Cat.nr.
▶	Greylag Goose <i>Anser anser</i>	0:07	Sunny Tseng	Lithuania	Zuvintas Biosphere	80	XC1152263
▶	Whooper Swan <i>Cygnus cygnus</i>	0:02	Sunny Tseng	Lithuania	Zuvintas Biosphere (lake shore)	80	XC1152320
▶	Mallard <i>Anas platyrhynchos</i>	0:06	Sunny Tseng	Lithuania	Nida (reed area)	0	XC1152613
▶	Mallard <i>Anas platyrhynchos</i>	0:10	Sunny Tseng	Lithuania	Vilnius forest	150	XC1151581

55 species, 204 recordings

16 new species recorded in Lithuania!

Autonomous recording units

2025 May - July

- 24 audio recording stations (24 hours)
- More than 10,000 hr of recordings

Great Cormorant colony

- Jokentë, biggest colony in LH
- ~3,800 breeding pairs, 14 ha
- Up to 50 nests in a single tree





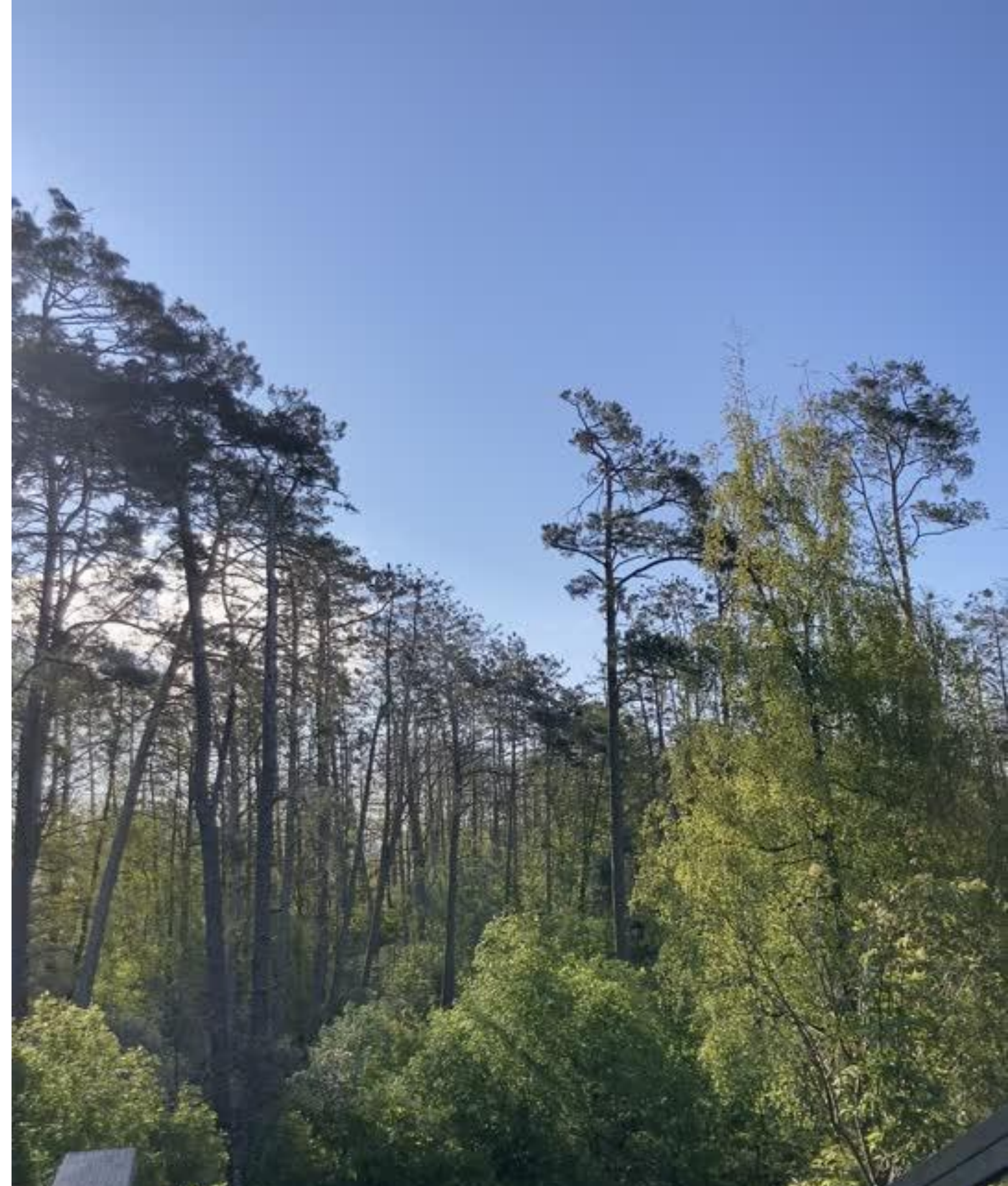
"curr-curr-curr"

"kirr-kirr-kirr"

"fhi-fhi-fhi"

"kra-wee...kra-wee.."

- Fish-eating bird
- Avian top predators
- Breeding in May - July



The acid forest

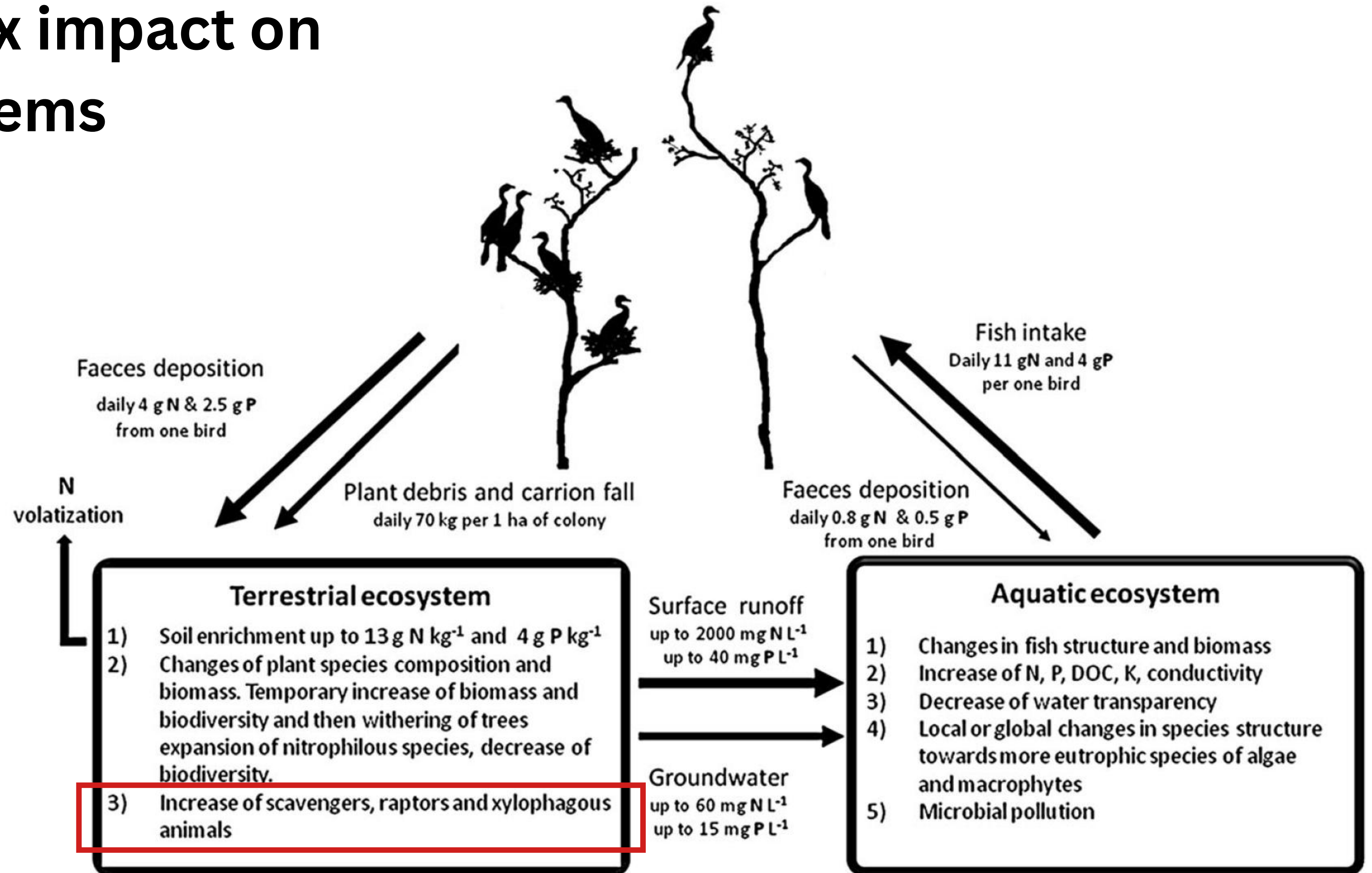
Conflicts between cormorants and fishermen
Cormorants' interaction with ecosystem?



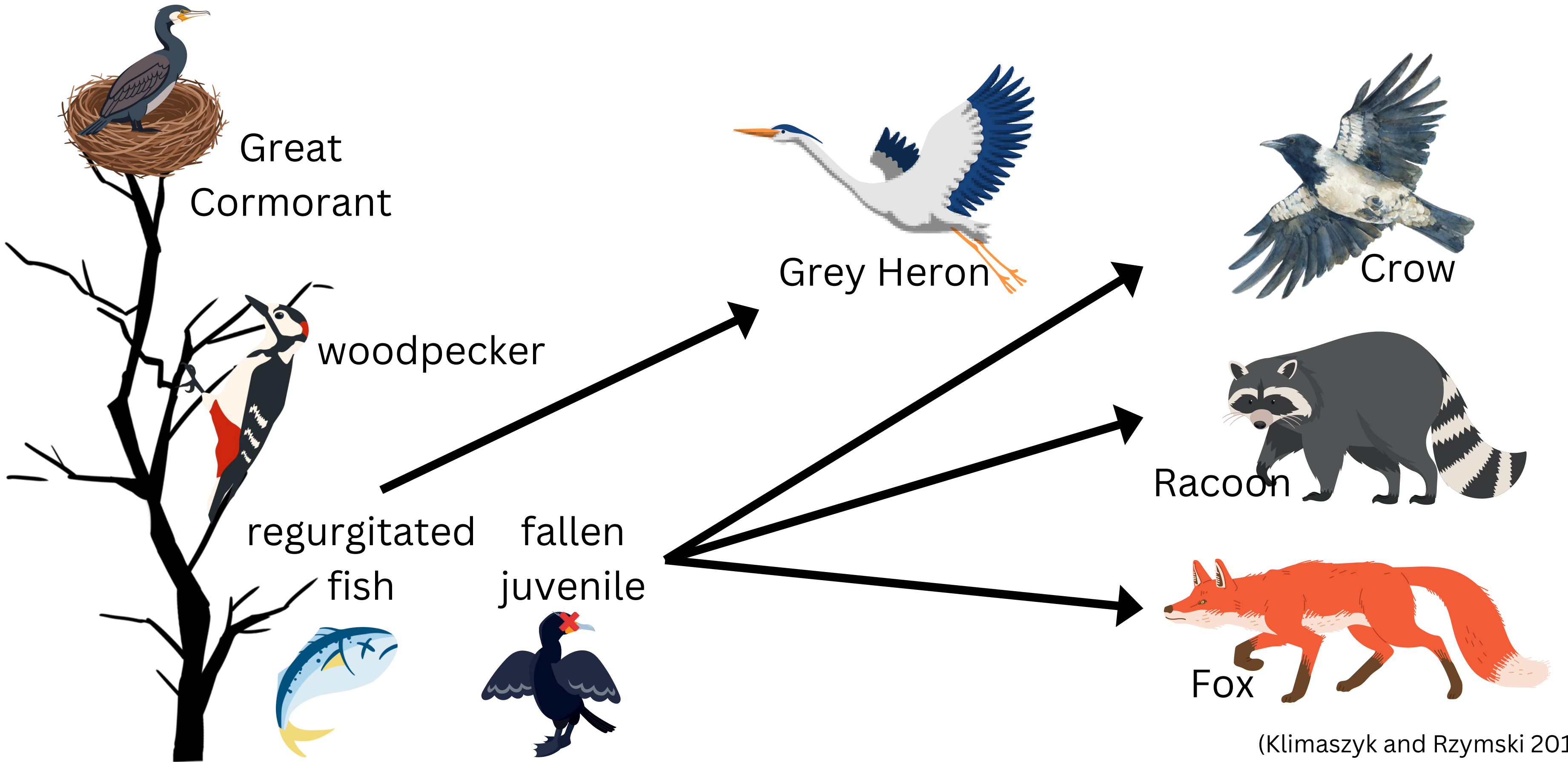
▲ nesting trees ▼ carrion



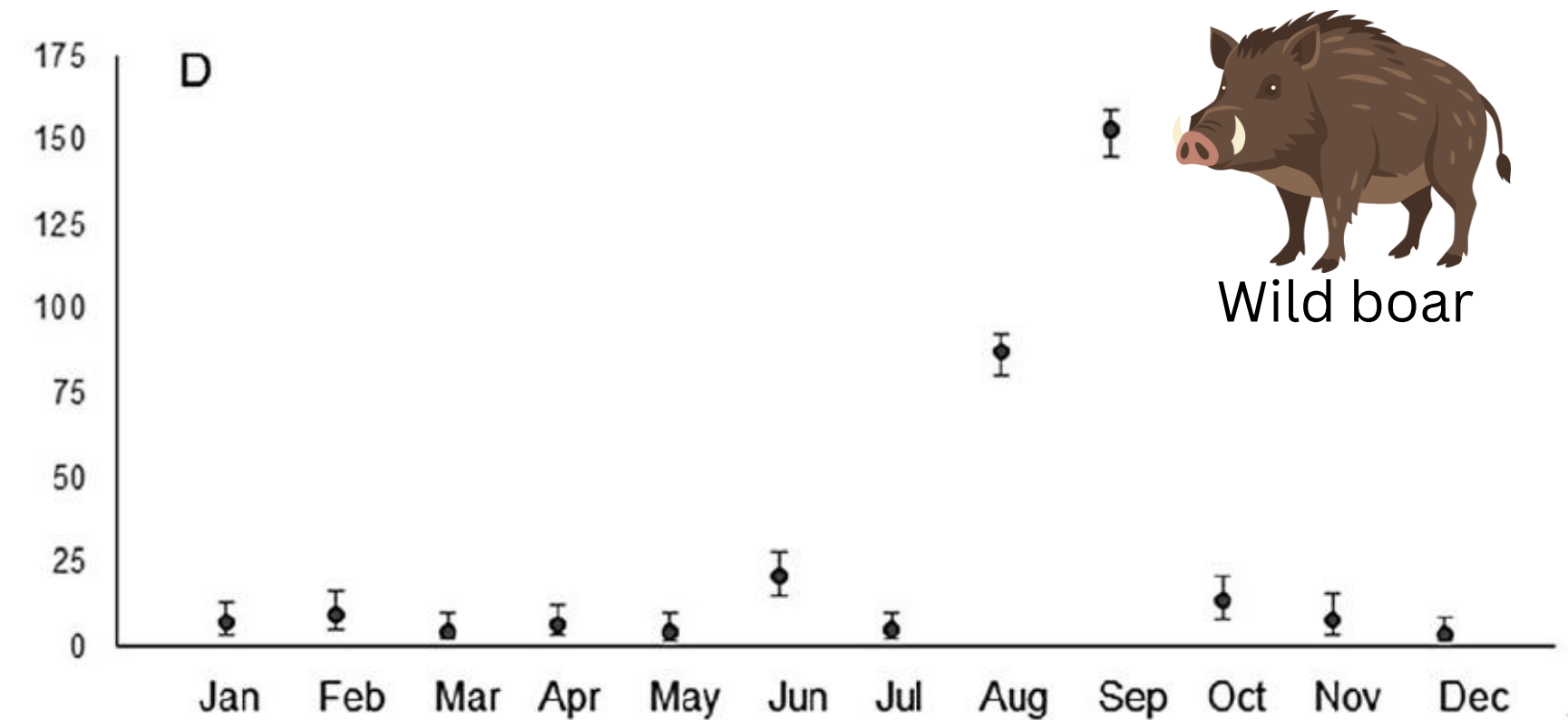
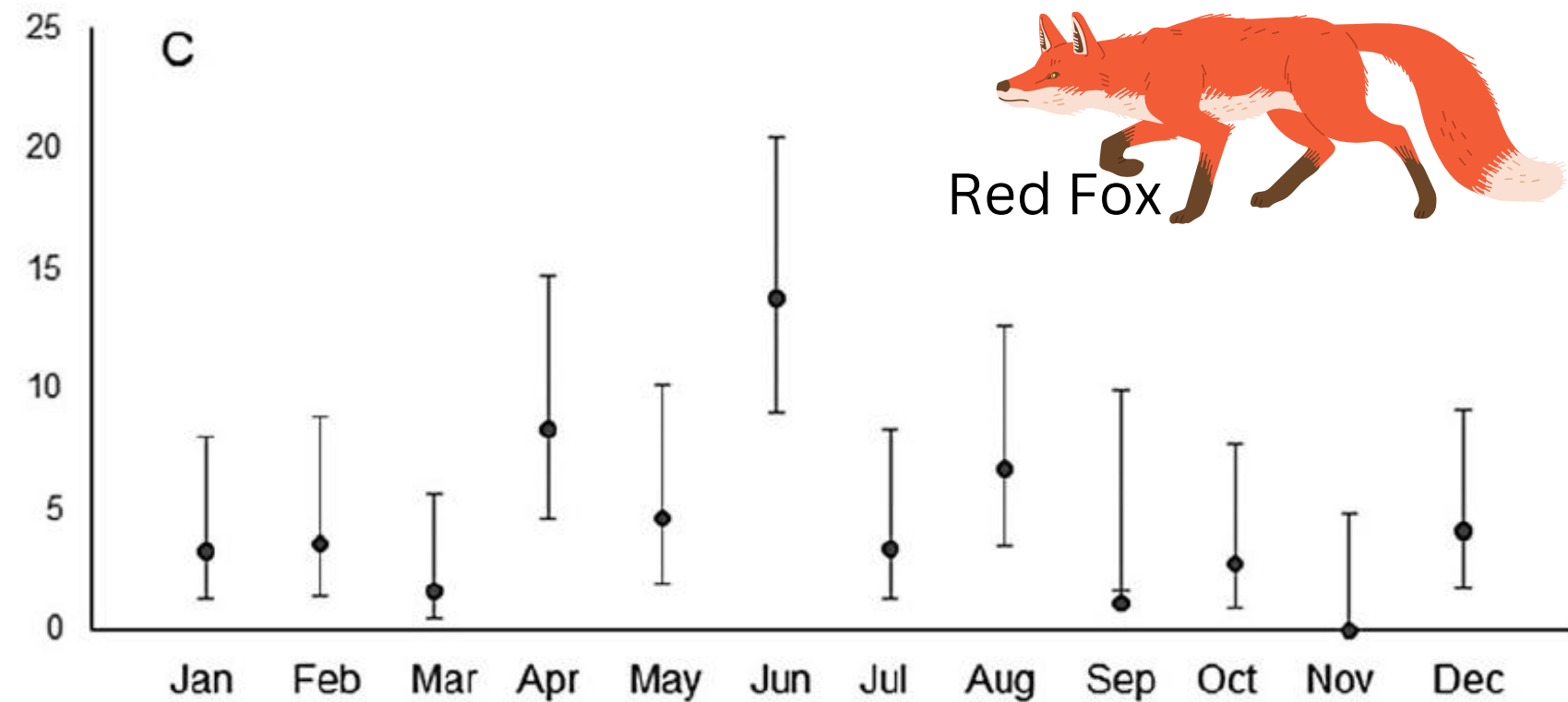
Complex impact on ecosystems



Bird and mammal's response

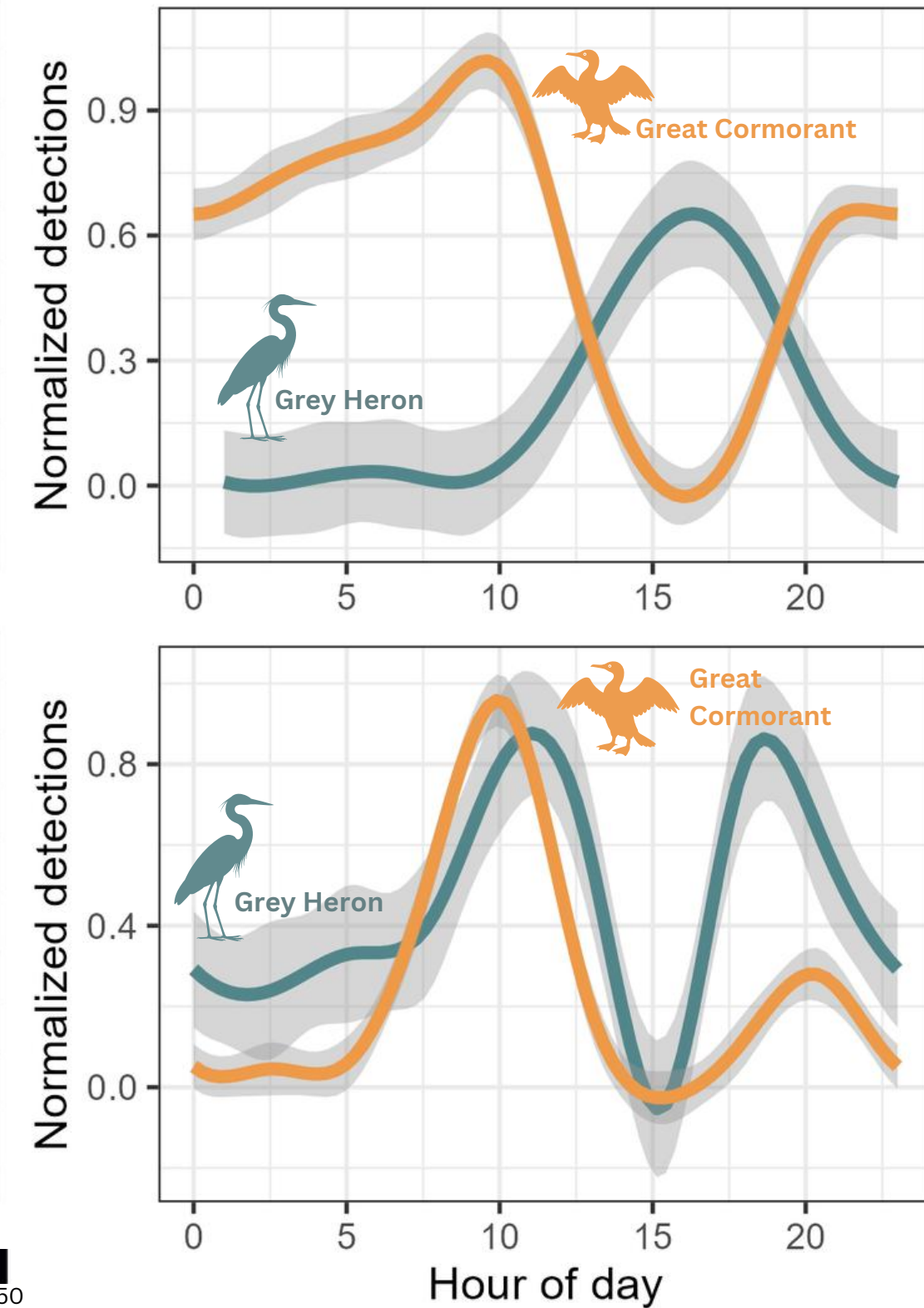
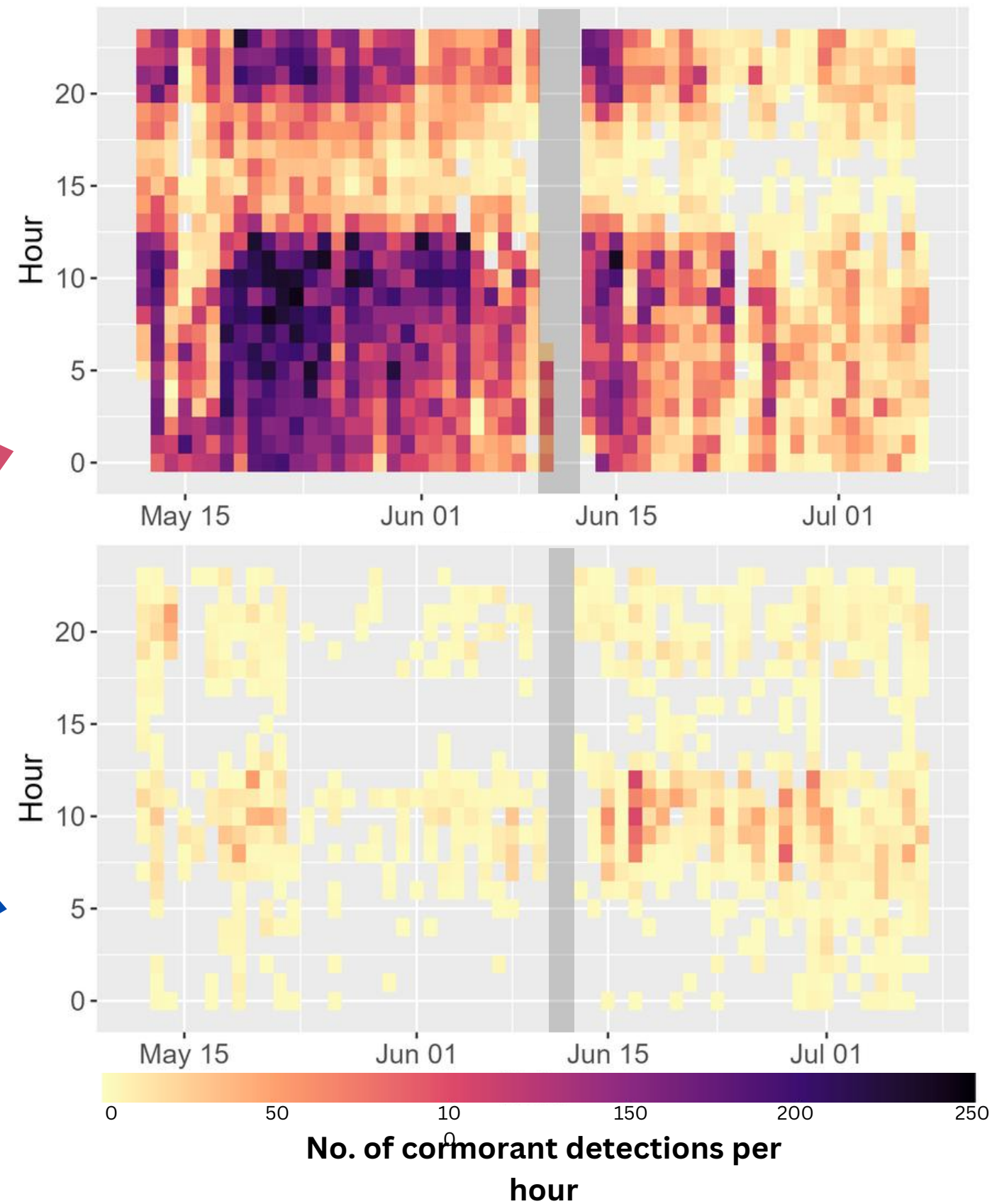
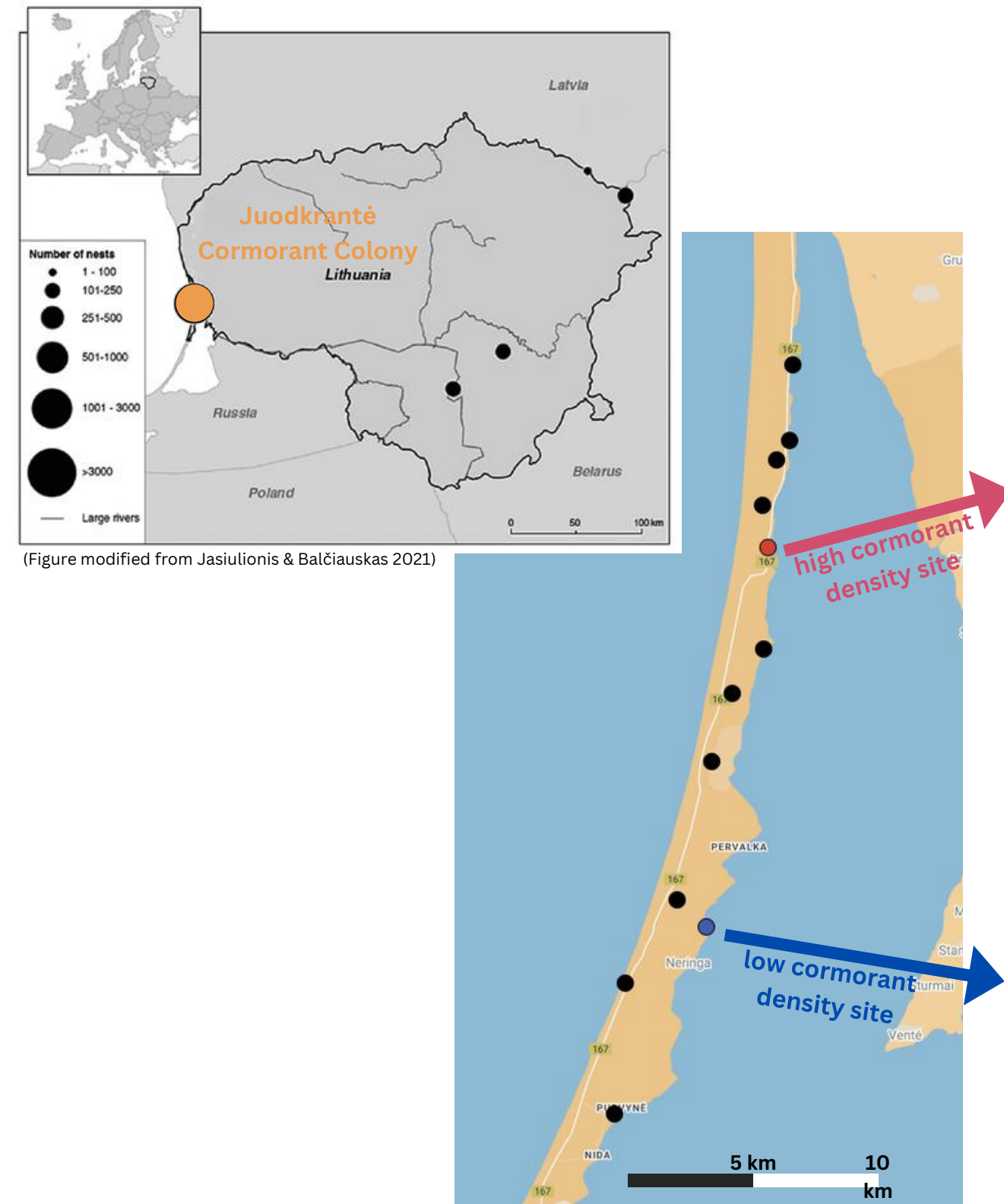


Camera trap for mammal's response



- Data from three camera sites
- No control site to reflect activity pattern without cormorant

What can we “hear” from the acoustic data?





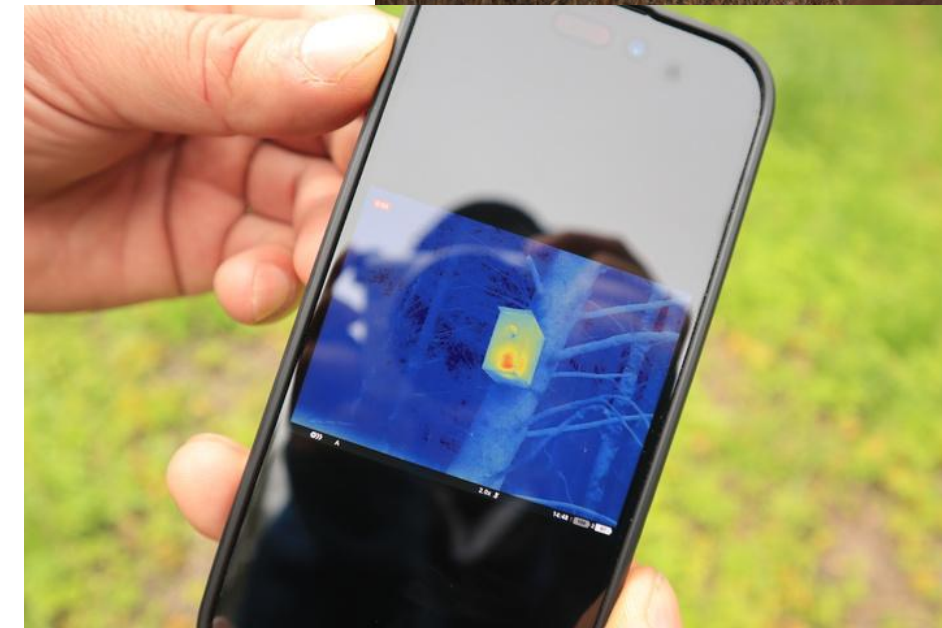
Sand dune ecosystem



Ventės Ragas ringing station



▲ mist net



◀ owl box



Acoustic monitoring in Lithuania

- Soundscape monitoring program in Lithuania
- Great Cormorant colony impact on local bird community
- Aquatic Warbler (endangered), Thrush Nightingale (vocal variation), etc.

Future directions

(aka. how I am hoping to continue)

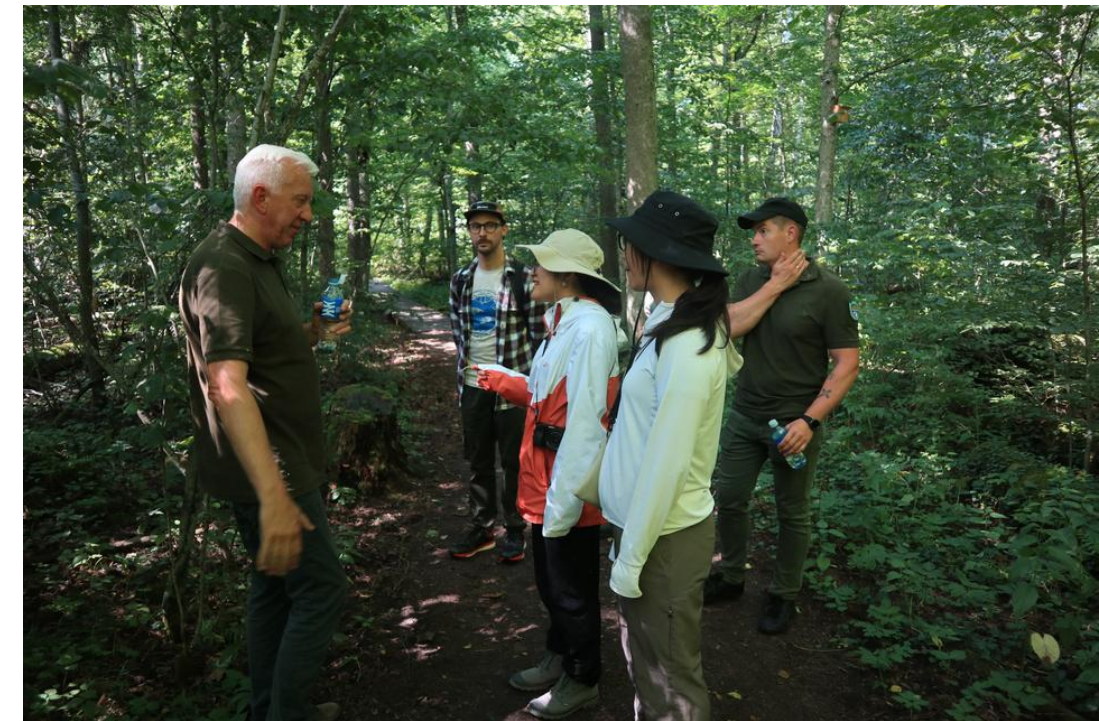
- Year-round, multiyear monitoring
 - Annual fieldwork
 - Local collaborators
- Collaborate with local academic research team



Taiwanese Representative Office



▲ with the Taiwanese ambassador in Lithuania



People, and culture

